

OPPORTUNITY RADAR

India-first opportunity intelligence platform for students & working professionals

Core idea: Don't build another random job board. Build a personalized system that tells people which opportunities are actually worth their time.

Students Internships • Hackathons • Research • Scholarships • Fellowships • Open Source • Women-focused opportunities	Working Professionals Opportunities matched to experience, skills, seniority, target role and company tier
Personalization Profile • Skills • Graduation year • Experience • Location • Role • Company tier • Opportunity type	ML layer Skill extraction • embeddings • semantic matching • ranking • behavioral personalization

Recommended MVP: Launch the opportunity discovery + personalization engine first. Add advanced ML, notifications, resume intelligence and professional career paths after real users start returning.

1. Product Vision

The product should become a daily/weekly career utility rather than a website users visit once. A user creates a profile and receives a feed containing only relevant, credible opportunities.

Opportunity categories

- Internships: SDE, MLE/AI, data, product, design, cybersecurity, finance/quant and other roles.
- Hackathons: college, national, company and open innovation hackathons.
- Research: IIT/IISc/lab research internships, professor-led opportunities and research programs.
- Scholarships: merit, need-based, women-focused, STEM and institution/company scholarships.
- Fellowships: technology, research, leadership and public-interest fellowships.
- Open Source: GSoC-style programs, mentorship programs, OSS internships and contribution opportunities.
- Women/Girl-focused: opportunities explicitly designed for women in CS/STEM, women returners and women-focused communities/programs.
- Professional opportunities: experienced-hire roles matched by years of experience, seniority, skills and target company tier.

Key product principle

Quality over quantity. The feed should not feel like 500 scraped links. Every opportunity should have source credibility, eligibility, deadline, company/institution quality and a match score.

2. User Segments

Student profile

- College / university and degree
- Year of study and expected graduation year
- Branch / specialization
- Skills, programming languages and tools
- Interests: SDE, ML, research, product, etc.
- Location preference: India, specific cities, remote, global
- Opportunity types and minimum preferences
- Women-focused opportunity preference (opt-in)

Working-professional profile

- Years of experience and current seniority
- Current role and target role
- Skills and technology stack
- Industry/domain
- Location / remote preference
- Target compensation range (optional)
- Preferred company tier
- Target companies / company blacklist

3. Company & Institution Tiering — Avoiding Random Opportunities

A major differentiator should be a quality/tier system. The platform should never imply that an opportunity is good merely because it was scraped.

Tier	Meaning	Examples / Signals
S / Elite	Highly selective, globally recognized	Major global tech, top HFT/quant, top research labs, highly selective programs
A / Strong	Strong established companies/institutions	Well-known product companies, major Indian tech firms, strong startups, leading universities
B / Good	Legitimate and useful opportunities	Verified startups, SMEs, established regional firms, credible labs
C / Emerging	Early-stage / lesser-known but potentially useful	Small startups, new organizations; require stronger verification

- Tier should be based on transparent signals: reputation, company size, funding/track record where appropriate, selectivity, source verification, institution quality and user feedback.
- Allow users to choose: Elite only / Elite + Strong / All verified.
- Never fabricate a tier. Use Unrated when evidence is insufficient.
- For working professionals, combine company tier with role seniority so a senior engineer does not receive entry-level noise.

4. Personalization & Filtering

This is the heart of the product. Users should be able to tune the feed instead of being forced into a generic list.

Filter	Examples
Opportunity type	Internship, research, hackathon, scholarship, fellowship, open source
Role	SDE, MLE, data science, research, product, design, cybersecurity
Experience	0 years, 1-2, 3-5, 5+ years
Company tier	S, A, B, C, unrated
Location	India, Bangalore, Hyderabad, Delhi, remote, global
Eligibility	Graduation year, degree, branch, experience, women-focused eligibility
Deadline	Today, 3 days, 1 week, custom

Example personalized feed

Profile: 2nd-year CSE student • graduating 2028 • Python/C++/ML • Bangalore/remote • SDE + MLE • wants strong companies.

Feed: Google SWE Intern — 94% match • IISc research internship — 91% • strong AI hackathon — 88% • women-in-tech program — 86%. A random low-quality marketing internship would be suppressed even if it contains the keyword “software”.

5. ML / Recommendation Architecture

Start with a hybrid system. You do not need a giant LLM to make the product intelligent.

Pipeline

Opportunity sources → ingestion → cleaning → deduplication → structured extraction → embeddings → eligibility rules → quality/tier score → personalized ranking → feed/notifications

- Skill extraction: identify Python, C++, React, PyTorch, SQL, etc. from descriptions.
- Semantic embeddings: represent user interests/profile and opportunity descriptions as vectors.
- Eligibility rules: hard constraints such as graduation year or experience range.
- Hybrid ranking: combine semantic similarity, skill overlap, eligibility, company tier, location, deadline urgency and user preferences.
- Explainability: show “Why this matches you?” with positive and missing signals.
- Behavioral learning later: clicks, saves, applies and ignores can become training signals for a learning-to-rank model.

6. Recommended Tech Stack

Layer	Recommended	Purpose
Frontend	Next.js + TypeScript + Tailwind + shadcn/ui	Dashboard, feed, profiles, filters, responsive UI
Backend	FastAPI + Python	REST API, auth integration, business logic, ML endpoints
Database	PostgreSQL via Supabase	Users, opportunities, profiles, saves, analytics
Vector search	pgvector	Semantic opportunity/user matching
ML	sentence-transformers + scikit-learn	Embeddings, similarity, ranking and classification
Ingestion	Python + Requests + BeautifulSoup + Playwright where permitted	Collect public opportunities from permitted sources
Scheduling	GitHub Actions / platform cron / lightweight worker	Refresh sources and process new opportunities
Auth	Supabase Auth	Email/Google login and sessions
Notifications	Email first; Telegram later	Daily/urgent opportunity alerts
Deployment	Vercel + a free/low-cost Python backend host	Public production deployment
Observability	Simple logs + product analytics	Track errors, usage and retention

7. Can It Be Free?

Yes for the MVP, if traffic and compute stay within the free tiers of the services you select. The architecture deliberately avoids paid LLM APIs at the beginning.

Component	MVP choice	Expected MVP cost
Frontend hosting	Vercel free tier	₹0 initially
Database/auth	Supabase free tier	₹0 initially
Embeddings	Open-source sentence-transformers	₹0 API cost
Vector DB	pgvector in PostgreSQL	₹0 extra
Code/CI	GitHub + Actions within limits	₹0 initially
Email	Free-tier email provider	₹0 initially within quota
Domain	.in or other custom domain	Usually paid; optional at launch
LLM API	Avoid initially; use rules + open-source models	₹0 initially

- Free tiers change over time, so verify current limits before launch.
- The first likely costs are a custom domain and infrastructure after traffic grows.
- Do not pay for GPU hosting initially. Generate embeddings offline/batch or on modest CPU infrastructure.

8. Core Data Model

users → account/auth information

profiles → education, experience, preferences, location, target roles

skills → canonical skill vocabulary

opportunities → title, organization, type, source, deadline, location, description, tier, verification

opportunity_skills → many-to-many skill mapping

saved_opportunities → saves and personal lists

applications → optional application tracking

interactions → impression, click, save, apply, ignore

notifications → email/Telegram delivery status

company_profiles → tier, industry, size signals, verification and aliases

9. Main Screens

#	Screen	Purpose
1	Landing page	Explain “relevant opportunities, not random listings.”
2	Onboarding	Education/experience + skills + target roles + company tier + location.
3	Personalized feed	Ranked opportunities with match %, deadline and tier.
4	Opportunity page	Eligibility, source, organization, why it matches, deadline, apply link.
5	Filters	Type, role, experience, tier, location, eligibility, deadline.
6	Saved / applied	Simple personal tracker.
7	Profile	Update preferences and skills.

#	Screen	Purpose
8	Daily digest	Top opportunities since the previous digest.

10. Build Roadmap

Phase 1 — MVP: Authentication → profile → opportunity database → verified sources → search/filter → personalized feed → deploy.

Phase 2 — Intelligence: Skill extraction → embeddings → hybrid matching → company tiering → “why this matches” → deduplication.

Phase 3 — Retention: Daily email → saved/applied tracker → deadline alerts → Telegram → weekly digest.

Phase 4 — Professional mode: Years of experience → seniority-aware matching → target-company tiers → role-level filtering → experienced-hire opportunities.

Phase 5 — Learning recommender: Use clicks/saves/applies to learn user preferences and improve ranking.

Phase 6 — Scale: More sources, source health monitoring, moderation, user reports, better ranking and analytics.

11. Resume Value

If the product gets real traction, the strongest resume story is not “built a job portal.” It is a production recommendation platform with real users.

Possible resume bullet after genuine traction

“Built and deployed an India-focused opportunity intelligence platform aggregating verified internships, research programs, hackathons, scholarships, fellowships and open-source opportunities; implemented hybrid semantic + rule-based personalization using sentence-transformer embeddings, pgvector and eligibility/ranking signals; added company-tier filtering, deadline alerts and behavioral analytics.”

Traction metrics to track

- Registered users
- Weekly/monthly active users
- Returning-user rate
- Opportunities indexed
- Verified sources
- Opportunity views
- Saves and apply clicks
- Daily digest open/click rate
- Recommendation click-through rate

12. What Makes This Different

Typical job board	Your platform
Shows many listings	Shows the most relevant listings first
Keyword search	Semantic + skill-based matching
Mostly jobs	Jobs + research + hackathons + scholarships + fellowships + OSS
One-size-fits-all	Student + professional profiles
Random companies	Transparent company/institution tiering
Passive browsing	Daily digest + deadline-driven alerts
No explanation	“Why this matches you”

13. Final Recommendation

Build Opportunity Radar first. Keep the initial scope narrow enough to launch, but design the database and recommendation engine so it can naturally grow into a larger career platform.

The strongest long-term positioning is:

“A personalized opportunity intelligence platform for students and professionals — not a random job board.”

Start with high-quality Indian opportunities, especially internships, research, scholarships, hackathons, women-focused programs, fellowships and open source. Then add experienced-hire matching based on years of experience and company tier.